

Calculus I Lesson 47

Formula A: $\int \frac{1}{\sqrt{a^2 - u^2}} du = \arcsin\left(\frac{u}{a}\right) + C$

Formula B: $\int \frac{1}{a^2 + u^2} du = \frac{1}{a} \arctan\left(\frac{u}{a}\right) + C$

Formula C: $\int \frac{1}{u\sqrt{u^2 - a^2}} du = \frac{1}{a} \operatorname{arcsec}\left(\frac{|u|}{a}\right) + C$

Question 1:

Find $\int \frac{dx}{\sqrt{25-x^2}}$

Hint: $\int \frac{1}{\sqrt{a^2-u^2}} du = \arcsin\left(\frac{u}{a}\right) + C$ $\int \frac{1}{a^2+u^2} du = \frac{1}{a} \arctan\left(\frac{u}{a}\right) + C$ $\int \frac{1}{u\sqrt{u^2-a^2}} du = \frac{1}{a} \operatorname{arcsec}\left(\frac{|u|}{a}\right) + C$

Let $u^2 = x^2 \Rightarrow u = x \Rightarrow x = u; \quad \frac{du}{dx} = 1 \Rightarrow du = dx$
 $a^2 = 25$

a) Write the integral $\int \frac{1}{\sqrt{25-x^2}} dx$ in terms of u : $\int \underline{\hspace{1cm} ? \hspace{1cm}} \cdot du$

b) $\int \frac{1}{\sqrt{25-x^2}} dx = \underline{\hspace{1cm} ? \hspace{1cm}}$

Question 2:

Find $\int \frac{1}{x\sqrt{9x^2-1}} dx$

Hint: $\int \frac{1}{\sqrt{a^2-u^2}} du = \arcsin\left(\frac{u}{a}\right) + C$ $\int \frac{1}{a^2+u^2} du = \frac{1}{a} \arctan\left(\frac{u}{a}\right) + C$

$$\int \frac{1}{u\sqrt{u^2-a^2}} du = \frac{1}{a} \operatorname{arcsec}\left(\frac{|u|}{a}\right) + C$$

Let $u^2 = 9x^2 \Rightarrow u = 3x \Rightarrow x = \frac{1}{3}u$ $\frac{du}{dx} = 3 \Rightarrow \frac{1}{3}du = dx$
 $a^2 = 1$

a) Write the integral $\int \frac{1}{x \cdot \sqrt{9x^2-1}} dx$ in terms of u : $\int \frac{1}{x \cdot \sqrt{9x^2-1}} dx = \int \frac{1}{\frac{1}{3}u \cdot \sqrt{u^2-1}} \cdot \frac{1}{3} du = \underline{\hspace{2cm}} ?$

b) $\int \frac{1}{x \cdot \sqrt{9x^2-1}} dx = \underline{\hspace{2cm}} ?$

Question 3:

Find $\int \frac{1}{\sqrt{1-(x+4)^2}} dx$

Hint: $\int \frac{1}{\sqrt{a^2-u^2}} du = \arcsin\left(\frac{u}{a}\right) + C$ $\int \frac{1}{a^2+u^2} du = \frac{1}{a} \arctan\left(\frac{u}{a}\right) + C$ $\int \frac{1}{u\sqrt{u^2-a^2}} du = \frac{1}{a} \operatorname{arcsec}\left(\frac{|u|}{a}\right) + C$

Let $u^2 = (x+4)^2 \Rightarrow u = x+4 \Rightarrow x = u-4; \quad \frac{du}{dx} = 1 \Rightarrow du = dx$
 $a^2 = 1$

a) Write the integral $\int \frac{1}{\sqrt{1-(x+4)^2}} dx$ in terms of u : $\int \underline{\hspace{1cm}} \cdot du$

b) $\int \frac{1}{\sqrt{1-(x+4)^2}} dx \underline{\hspace{1cm}} ?$

Question 4:

Find $\int \frac{1}{\sqrt{1-25x^2}} dx$

Hint: $\int \frac{1}{\sqrt{a^2 - u^2}} du = \arcsin\left(\frac{u}{a}\right) + C$

$$\int \frac{1}{a^2 + u^2} du = \frac{1}{a} \arctan\left(\frac{u}{a}\right) + C$$

$$\int \frac{1}{u\sqrt{u^2 - a^2}} du = \frac{1}{a} \operatorname{arcsec}\left(\frac{|u|}{a}\right) + C$$

$$\begin{aligned} \text{Let } u^2 = 25x^2 &\Rightarrow u = 5x \Rightarrow x = \frac{1}{5}u; & \frac{du}{dx} = 5 &\Rightarrow \frac{1}{5} du = dx \\ a^2 = 1 & \end{aligned}$$

a) Write the integral $\int \frac{1}{\sqrt{1-25x^2}} dx$ in terms of u : $\int \frac{1}{\sqrt{1-u^2}} \cdot \frac{1}{5} \cdot du = \frac{1}{5} \cdot \int \underline{\hspace{1cm}} \cdot du$

b) $\int \frac{1}{\sqrt{1-25x^2}} dx = \underline{\hspace{1cm}}$

Question 5:

Find $\int \frac{1}{\sqrt{4+5x^2}} dx$

Hint: $\int \frac{1}{\sqrt{a^2 - u^2}} du = \arcsin\left(\frac{u}{a}\right) + C$

$$\int \frac{1}{a^2 + u^2} du = \frac{1}{a} \arctan\left(\frac{u}{a}\right) + C$$

$$\int \frac{1}{u\sqrt{u^2 - a^2}} du = \frac{1}{a} \operatorname{arcsec}\left(\frac{|u|}{a}\right) + C$$

$$\begin{array}{l} \text{Let } u^2 = 5x^2 \Rightarrow u = \sqrt{5}x \Rightarrow x = \frac{1}{\sqrt{5}}u; \\ a^2 = 4 \end{array} \quad \frac{du}{dx} = \sqrt{5} \Rightarrow \frac{1}{\sqrt{5}} du = dx$$

a) Write the integral $\int \frac{1}{\sqrt{4+5x^2}} dx$ in terms of u : $\int \frac{1}{\sqrt{4+5x^2}} dx = \int \frac{1}{\sqrt{a^2 + u^2}} \cdot \frac{1}{\sqrt{5}} du = \frac{1}{\sqrt{5}} \cdot \int \underline{\hspace{1cm}} \cdot du$

b) $\int \frac{1}{\sqrt{4+5x^2}} dx = \underline{\hspace{1cm}} ?$

Question 6:

Find $\int \frac{1}{\sqrt{10+9x^2}} dx$

Hint: $\int \frac{1}{\sqrt{a^2 - u^2}} du = \arcsin\left(\frac{u}{a}\right) + C$

$\int \frac{1}{a^2 + u^2} du = \frac{1}{a} \arctan\left(\frac{u}{a}\right) + C$

$\int \frac{1}{u\sqrt{u^2 - a^2}} du = \frac{1}{a} \operatorname{arcsec}\left(\frac{|u|}{a}\right) + C$

Let $u^2 = 9x^2 \Rightarrow u = 3x \Rightarrow x = \frac{1}{3}u;$ $\frac{du}{dx} = 3 \Rightarrow \frac{1}{3}du = dx$
 $a^2 = 10$

a) Write the integral $\int \frac{1}{\sqrt{10+9x^2}} dx$ in terms of u : $\int \frac{1}{\sqrt{10+9x^2}} dx = \int \frac{1}{\sqrt{a^2 + u^2}} \cdot \frac{1}{3} du = \frac{1}{3} \cdot \int \underline{\hspace{1cm}} \cdot du$

b) $\int \frac{1}{\sqrt{10+9x^2}} dx = \underline{\hspace{1cm}} ?$

Question 7:

Find $\int \frac{4x+13}{x^2+1} dx = \int \frac{4x}{x^2+1} dx + \int \frac{13}{x^2+1} dx$

a) $\int \frac{4x}{x^2+1} dx = \underline{\hspace{2cm}} ?$

Hint: Let $u = x^2 + 1$; use substitution method.

b) $\int \frac{13}{x^2+1} dx = 13 \int \frac{1}{x^2+1} dx = \underline{\hspace{2cm}} ?$

Hint: $\int \frac{1}{a^2+u^2} du = \frac{1}{a} \arctan\left(\frac{u}{a}\right) + C$

c) $\int \frac{4x+13}{x^2+1} dx = \underline{\hspace{2cm}} ?$

Question 8:

$$\text{Find } \int \frac{2x+1}{x^2-16} dx = \int \frac{2x+1}{x^2-16} dx + \int \frac{1}{x^2-16} dx$$

$$\text{a) } \int \frac{2x}{x^2-16} dx = 2 \cdot \int \frac{1}{x^2-16} \cdot x \cdot dx = \underline{\hspace{2cm}}$$

Hint: Let $u = x^2-16$; use substitution method.

$$\text{b) } \int \frac{1}{x^2-16} dx = \int \frac{1}{x^2-16} dx \quad \text{Hint: } \frac{1}{x^2-16} = \frac{1}{8(x-4)} - \frac{1}{8(x+4)}; \quad \int \frac{1}{x-a} = \ln|x-a| + C$$

Question 9:

Find $\int \frac{1}{4x^2 - 1} dx$

Hint: $\frac{1}{4x^2 - 1} = \frac{1}{2(2x - 1)} - \frac{1}{2(2x + 1)}$; $\int \frac{1}{ax \pm b} = \frac{1}{a} \ln|x \pm b| + C$

a) $\int \frac{1}{2(2x - 1)} dx = \underline{\hspace{2cm}} ?$

b) $\int \frac{1}{2(2x + 1)} dx = \underline{\hspace{2cm}} ?$

c) $\int \frac{1}{4x^2 - 1} dx = \underline{\hspace{2cm}} ?$

Question 10:

Find $\int \frac{1}{9x^2 - 16} dx$

Hint: $\frac{1}{9x^2 - 16} = \frac{1}{8(3x - 4)} - \frac{1}{8(3x + 4)}$; $\int \frac{1}{ax \pm b} = \frac{1}{a} \ln|x \pm b| + C$

a) $\int \frac{1}{8(3x - 4)} dx = \underline{\hspace{2cm}}?$

b) $\int \frac{1}{8(3x + 4)} dx = \underline{\hspace{2cm}}?$

c) $\int \frac{1}{9x^2 - 16} dx = \underline{\hspace{2cm}}?$

